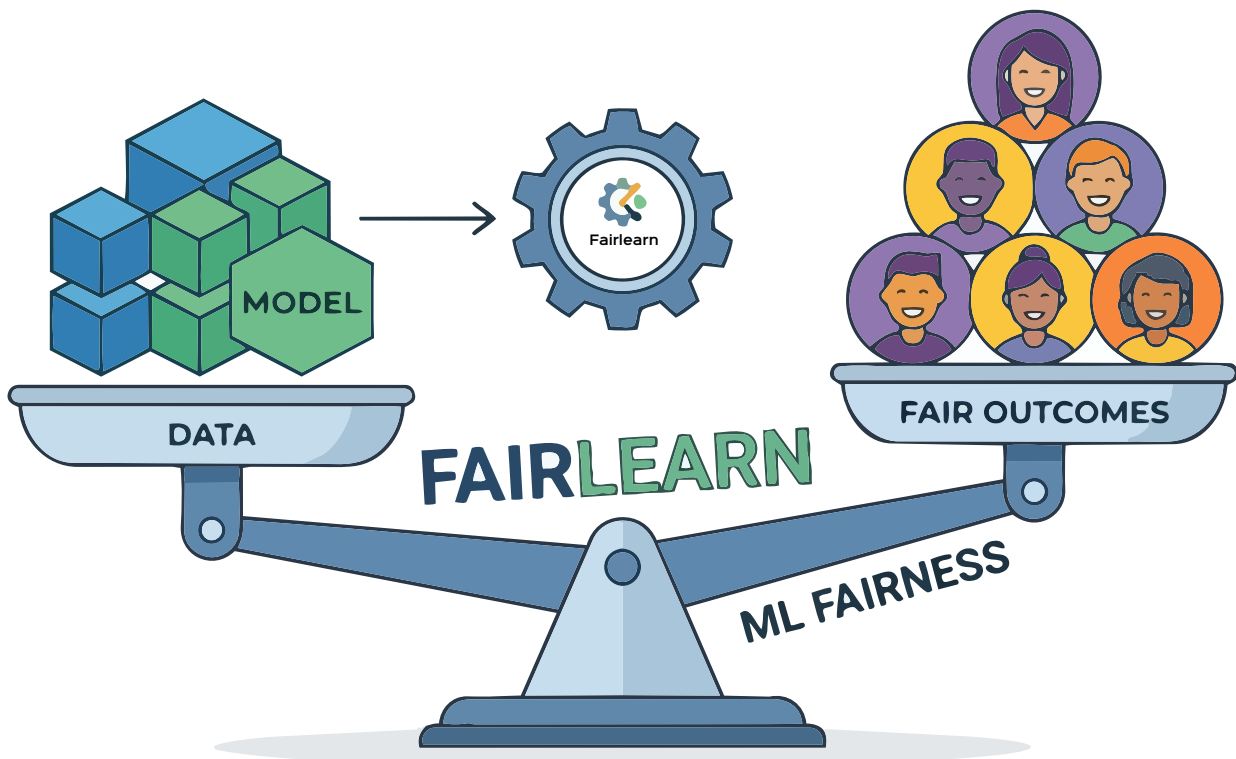


Ensuring Fairness in AI with Fairlearn

One of the aims of responsible AI is to make sure no injustice creeps into the data that large language models are based on. Fairlearn is an open source tool that can help with this.



The amount of data that is being used to train large language models today, which have billions of parameters, causes serious privacy and other concerns. Responsible AI and AI governance have become very important and relevant today. The fundamentals of responsible AI include transparency, bias, fairness, privacy, and accountability. Here, we will primarily focus on fairness.

Fairness is basically the practice of developing or using algorithms and models in such a way that they provide decisions that are equitable, unbiased

and non-discriminatory to different demographic groups.

Fairlearn is a tool that helps us ensure and improve fairness while building a model. It is an open source Python toolkit, initially developed by Microsoft, and is designed to measure and mitigate unfairness in ML models. It provides easy-to-use tools for evaluating fairness and adjusting models to reduce bias.

Ensuring fairness in a model requires two steps. The first is to measure fairness and the second is to mitigate the unfairness. Fairlearn

helps us with both! To measure fairness, it first identifies disparities in metrics like accuracy or false-positive rate across sensitive groups (e.g., gender, race). Then, to mitigate the unfairness it modifies models or predictions to reduce bias while balancing performance.

Fairlearn supports three approaches to fairness. The first is during pre-processing, where it helps adjust the data before training to remove bias. During processing, it helps train the models with fairness constraints, and